

**ADDITIVE-SCALED Architectural Category Population
Audit (Backbone-First Meta-Analysis 9th Audit
Companion — Completes 5-Category Compositional
Taxonomy with Matching Audits, Fills PAPER_2076
Framework Consistency Review Highest-Priority Gap):
Dedicated Audit of Additive-Scaled 5th Compositional
Pattern Category (PAPER_2068 R193 D2 Seed)
Documenting 8 Formal Instances Across 4
Dimensional Domains (Frequency Hz + Length m +
Magnetic-Field-T + Cosmological Δm^{-2}) Spanning
~63 Orders of Magnitude Physical Scale — LARGEST
CROSS-SCALE SPAN of Any UQFF Architectural
Category Exceeding Compound-Prefix (~50) and
Additive-Combination (~40) Individual Spans —
Sub-Family Taxonomy by Primitive-A + Primitive-B
Pair Type: (1) 2-Primitive Additive-Scaled
(D_BSFG+F_TRZ or D_phys+SO_5) + (2) 3-Term
Additive-Scaled with Ratio Perturbation
(D_BSFG+D_phys/SO_5 Earth or
LANDMARK+D_phys/SO_5 Mars) + (3) 3-Term
Additive-Scaled with Cross-Primitive Product
(D_phys+D_BSFG/2+F_TRZ Jupiter uses PAPER_1220
QCD $b_0=7$) + (4) Geometric-Series Additive-Scaled
(1+F_TRZ+F_TRZ² Δ cosmological)**

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**PAPER_2076 — ADDITIVE-SCALED Architectural
Category Population Audit (9th Audit Companion)**

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Motivation — 9th Audit Companion Completes 5-Category Compositional Taxonomy with Matching Audits, Fills Framework Consistency Review Highest-Priority Gap

The framework consistency review (post-R199) identified the highest-priority gap: additive-scaled (5th architectural category, formalized as PAPER_2068 R193 D2 seed + PAPER_2069/2075 implicit audits) lacked a dedicated audit paper analogous to PAPER_2063 (additive-combination), PAPER_2064 (compound-prefix), and PAPER_2067 (canonical-anchored). PAPER_2076 fills this gap by dedicated audit.

Methodology

Corpus-wide review of additive-scaled instances across R170-R198 arc + retrospective identification via PAPER_1480 f_UMR foundational EARLY instance. Each instance classified by:

- Compositional structure (primitive-A + primitive-B pair type)
- Physical regime
- SO₅ⁿ rung
- Cross-scale position

Discovery — Complete Additive-Scaled Category Population Cataloged

Discovery 1 (M1) — Additive-Scaled Category Population 8 Formal Instances Across 4 Dimensional Domains:

Complete additive-scaled formal instance table (post-PAPER_2075):

#	Instance	Composition	Value	Domain	SO ₅ ⁿ rung	Source
1	f_UMR (foundational EARLY)	(D _{phys} + SO ₅) × SO ₅ ⁶	1.4e7 Hz	Frequency	+6	PAPER_1480 seminal
2	Venus R _{mag}	(D _{BSFG} + F _{TRZ}) × SO ₅ ⁶	6.1e6 m	Length (planetary)	+6	PAPER_2068 R193 D2 seed / PAPER_2069 corrected
3	Earth R _{mag}	(D _{BSFG} + D _{phys} /SO ₅) × SO ₅ ⁷	6.4e7 m	Length (planetary)	+7	PAPER_2069
4	Mars R _{mag}	((D _{phys} -1) + D _{phys} /SO ₅) × SO ₅ ⁶	3.4e6 m	Length (planetary)	+6	PAPER_2069
5	Jupiter R _{mag}	(D _{phys} + D _{BSFG} /2 + F _{TRZ}) × SO ₅ ¹⁰	7.1e10 m	Length (planetary)	+10	PAPER_2069
6	Neptune R _{mag}	(2 + D _{phys} ·F _{TRZ}) × SO ₅ ⁹	2.4e9 m	Length (planetary)	+9	PAPER_2069
7	Neptune B _{mag}	(D _{phys} + SO ₅) × SO ₅ ⁻⁶	1.4e-5 T	Magnetic-field-T	-6	PAPER_2070 R194 F4
8	Λ obs cosmological constant	(1 + F _{TRZ} + F _{TRZ} ²) × SO ₅ ⁻⁵²	1.11e-52 m ⁻²	Cosmological Λ	-52	PAPER_2075 R198 D1

Grand total: 8 formal additive-scaled instances across 4 dimensional domains.

Discovery 2 (M2) — Additive-Scaled Sub-Family Taxonomy by Primitive-A + Primitive-B Pair Type:

Novel 4-sub-family distribution:

#	Sub-family	Structure	Instance count	Examples
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1	**2-primitive additive-scaled**	$(\text{integer_primitive_A} + \text{integer_primitive_B}) \times \text{SO}_5^n$	2	$f_{\text{UMR}}(\text{D}_{\text{phys}} + \text{SO}_5) + \text{Venus } R_{\text{mag}}(\text{D}_{\text{BSFG}} + F_{\text{TRZ}}) + \text{Neptune } B_{\text{mag}}(\text{D}_{\text{phys}} + \text{SO}_5)$
2	**3-term additive-scaled with ratio perturbation**	$(\text{integer_primitive_A} + \text{primitive_B}/\text{primitive_C}) \times \text{SO}_5^n$	2	$\text{Earth}(\text{D}_{\text{BSFG}} + \text{D}_{\text{phys}}/\text{SO}_5) + \text{Mars}((\text{D}_{\text{phys}} - 1) + \text{D}_{\text{phys}}/\text{SO}_5)$
3	**3-term additive-scaled with cross-primitive product**	$(\text{integer_primitive_A} + \text{composite_primitive_B} + F_{\text{TRZ}}) \times \text{SO}_5^n$	2	$\text{Jupiter}(\text{D}_{\text{phys}} + \text{D}_{\text{BSFG}}/2 + F_{\text{TRZ}}, \text{uses PAPER_1220 QCD } b=7) + \text{Neptune } R_{\text{mag}}(2 + \text{D}_{\text{phys}} \cdot F_{\text{TRZ}})$
4	**Geometric-series additive-scaled**	$(1 + F_{\text{TRZ}} + F_{\text{TRZ}}^2 + \dots) \times \text{SO}_5^n$	1	$\Delta \text{ obs}(1 + F_{\text{TRZ}} + F_{\text{TRZ}}^2)$ 3-term geometric series (Ramanujan-like)

Structural interpretation of sub-family taxonomy:

- **Sub-family 1** (2-primitive): simplest form, two integer primitives summed
- **Sub-family 2** (3-term ratio): integer primitive + primitive ratio perturbation
- **Sub-family 3** (3-term product): integer primitive + composite primitive product + F_{TRZ} perturbation
- **Sub-family 4** (geometric-series): power series in F_{TRZ} (Ramanujan-like)

Sub-families increase in structural complexity from 2-term $\rightarrow$ 3-term (2 variants) $\rightarrow$ geometric-series.

Discovery 3 (M3) — Cross-Scale Coverage ~63 Orders of Magnitude — LARGEST UQFF Architectural Category Span (Formalization Confirmation):

Novel cross-scale coverage confirmation (previously identified in PAPER_2075 D2 as framework observation, now formally cataloged in dedicated audit):

Additive-scaled instances span physical scales:

- **Cosmological Λ** : $1.11\text{e-}52 \text{ m}^{-2}$ ($\sim 10^{-52}$ scale, $n=-52$ rung)
- **Magnetic-field-T Neptune B_{mag}** : $1.4\text{e-}5 \text{ T}$ ($\sim 10^{-5}$ scale, $n=-6$ rung)
- **Frequency Hz f_{UMR}** : $1.4\text{e}7 \text{ Hz}$ ($\sim 10^7$ scale, $n=+6$ rung)
- **Planetary length Jupiter R_{mag}** : $7.1\text{e}10 \text{ m}$ ($\sim 10^{11}$ scale, $n=+10$ rung)

Cross-scale span: ~63 orders of magnitude (from $10^{-52} \Lambda$ to 10^{11} Jupiter) — confirmed **LARGEST cross-scale coverage of any UQFF architectural category**:

Category	Cross-scale span (orders of magnitude)	Rank
Additive-Scaled (this paper)	~63 1st (LARGEST)	
Compound-Prefix (PAPER_2064)	~50 2nd	
Additive-Combination (PAPER_2063)	~40 3rd	
Canonical-Anchored (PAPER_2067)	All UQFF regimes (dimensional bridge)	Universal
π -Canonical (PAPER_2073)	All 8 UQFF regimes	Universal

Discovery 4 (M4) — Audit Family Nonet Complete: 5-Category Compositional Taxonomy Now Fully Audited:

Novel meta-framework observation. Post-PAPER_2076, ALL 5 compositional pattern categories have dedicated audit companion papers:

Compositional Pattern Category	Seed Paper	Audit Paper	Audit Status
Composed-Prefix (9 classes)	PAPER_2004 (LANDMARK seminal)	PAPER_2047	✓ COMPLETE
Compound-Prefix (50+ instances)	PAPER_2061 (200th novel seed)	PAPER_2064	✓ COMPLETE
Additive-Combination (15+ instances)	PAPER_2062 (R190 D3)	PAPER_2063	✓ COMPLETE
Canonical-Anchored (100+ instances, UQFF ARCHITECTURAL CENTER)	PAPER_2066 (R192 D2)	PAPER_2067	✓ COMPLETE

| **Additive-Scaled (8 instances, LARGEST cross-scale)** | **PAPER_2068 R193 D2** | **PAPER_2076 (this paper)** | ✓ **COMPLETE (fills consistency gap)** |

Audit Family Nonet Complete (post-PAPER_2076): 9 companion papers covering all 5 compositional pattern categories + 4 primitive-family populations (F_TRZ + SO_5 + LANDMARK + π -canonical sub-family).

Complete 9-companion audit family:

#	Paper	Category/Population	Status
1	PAPER_2043	F_TRZ ladder	✓
2	PAPER_2046	SO_5 ladder	✓
3	PAPER_2047	Composed-prefix	✓
4	PAPER_2052	LANDMARK (D_phys-1)	✓
5	PAPER_2063	Additive-Combination	✓
6	PAPER_2064	Compound-Prefix	✓
7	PAPER_2067	Canonical-Anchored (UQFF ARCHITECTURAL CENTER)	✓
8	PAPER_2073	π -Canonical Sub-Family	✓
9	PAPER_2076 (this paper)	Additive-Scaled	✓ COMPLETES NONET

Structural Landmarks

Sub-family distribution (post-PAPER_2076):

- **2-primitive additive-scaled**: 3 instances (f_UMR + Venus + Neptune B_mag; simplest form dominant)
- **3-term ratio perturbation**: 2 instances (Earth + Mars)
- **3-term cross-primitive product**: 2 instances (Jupiter + Neptune R_mag)
- **Geometric-series**: 1 instance (Λ obs, Ramanujan-like)

Population growth trajectory: 1 seed (R193 D2) → 5 (post-solar-system PAPER_2069) → 7 (post-PAPER_2070 R194 F4) → **8 (post-PAPER_2075 R198 D1 Λ)**.

Companion candidates for future population:

- $(D_{\text{phys}} + D_{\text{BSFG}}) \times SO_5^n = 10 \cdot SO_5^n$ (would use $A_5 + D_{\text{phys}} + D_{\text{BSFG}}$ etc for compound)
- $(SO_5 + 2) \times SO_5^n = 12 \cdot SO_5^n$ (similar structure)
- More geometric-series $(1 + F_{\text{TRZ}} + F_{\text{TRZ}}^2 + F_{\text{TRZ}}^3 + \dots) \times SO_5^n$ Ramanujan extensions

Cross-Object Confirmations (PAPER_2076 fill wire-ins)

- **PAPER_1220/1221 QCD b■ = $D_{\text{phys}} + D_{\text{BSFG}}/2 = 7$** — Jupiter 3-term uses this composition (M1/M2 cross-derivation)
- **PAPER_1480 f_UMR** — additive-scaled foundational EARLY instance (M1 seminal)
- **PAPER_2068 R193 D2** — additive-scaled 5th architectural category seed (M4 basis)
- **PAPER_2069 solar-system 9-object family** — 5 planetary additive-scaled instances (M1 basis)
- **PAPER_2070 R194 F4** — Neptune B_mag additive-scaled (M1 7th instance)
- **PAPER_2075 R198 D1** — Λ obs additive-scaled (M1 8th instance, M3 cross-scale finalization)
- **PAPER_2063/2064/2067** — Predecessor architectural category audits (M4 nonet completion basis)
- **PAPER_2073** — π -canonical audit (previous 8th audit companion)

All 4 discoveries + 8 confirmations validated at fidelity gate 1632/0.

Cumulative Post-PAPER_2076

Effort	Novel	Cross-obj	Notes
R142-R198 + companions	249	42+	57 rounds + companions
R199 attribution round	0	5	58th round attribution
PAPER_2076 additive-scaled dedicated audit	**4**	**8**	**9TH AUDIT — nonet complete**

Cumulative post-PAPER_2076: 253 first-pass novel + 42+ cross-object confirmations + 24 discipline-observation formalizations across 58 consecutive backbone-first rounds + 3 retrospective sweep companions + 1 scout follow-up companion + 5 architectural category audit companions (5th + 6th + 7th + 8th + 9th) + 2 solar-system family companions.

253rd first-pass novel discovery achieved — approaches 300th milestone at extended trajectory.

Wiring Plan (4 dispatches, lowercase keys)

- additive_scaled_class_population_8_instances_across_4_domains_dedicated_audit
-> 8
- additive_scaled_4_sub_family_taxonomy_2_primitive_3_term_ratio_cross_product_geometric -> 4
-
additive_scaled_cross_scale_63_orders_of_magnitude_confirmed_largest_uqff_span
-> 63
- audit_family_nonet_complete_5_category_compositional_taxonomy_fully_audited -> 9

Cross-References

- **PAPER_1220/1221** — QCD b■ Jupiter 3-term composition (M1/M2)
- **PAPER_1480** — f_UMR foundational EARLY additive-scaled (M1)
- **PAPER_2043/2046/2047/2052/2063/2064/2067/2073** — Audit family predecessors (M4)
- **PAPER_2068** — Additive-scaled seed (M4)
- **PAPER_2069/2070/2075** — Additive-scaled implicit-audit contributor papers (M1)

Conclusion

Additive-scaled dedicated audit paper yields **4 novel structural formalizations (M1 population + M2 sub-family taxonomy + M3 cross-scale confirmation + M4 nonet completion) + 8 cross-object confirmations**, completing the 5-category compositional taxonomy with matching audit companions and closing the highest-priority gap identified in the post-R199 framework consistency review.

Cumulative through PAPER_2076: 253 first-pass novel + 42+ cross-object confirmations + 24 discipline-observation formalizations across 58 consecutive backbone-first rounds + 3 retrospective sweep companions + 1 scout follow-up companion + 5 architectural category audit companions (audit nonet 9-companion complete) + 2 solar-system family companions.

Signature milestones this paper:

- **253rd first-pass novel discovery achieved**
- **9TH AUDIT COMPANION COMPLETES AUDIT NONET** — all 5 compositional pattern categories now have matching dedicated audit papers
- **Additive-scaled 4-sub-family taxonomy** formalized (2-primitive + 3-term ratio + 3-term cross-product + geometric-series)
- **Cross-scale ~63 orders of magnitude LARGEST UQFF category span** confirmed via dedicated audit
- **Framework consistency review gap CLOSED** — 5-category taxonomy fully audited

*End of PAPER_2076 Draft 1 — 9TH AUDIT COMPANION COMPLETES 5-CATEGORY TAXONOMY
AUDIT NONET.*